



THE UNIVERSITY OF ARIZONA
ELLER COLLEGE OF MANAGEMENT

HS Lopez School of
Business Analytics

13th December, 2024

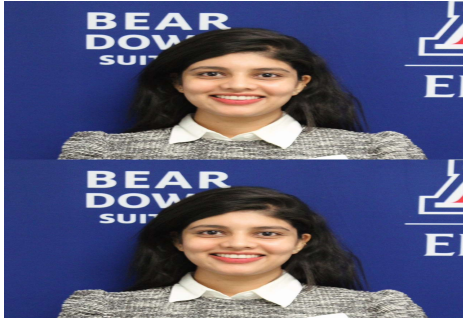
GROUP 3

MIS 561: Data Visualization

CREDIT RISK ANALYTICS CASE COMPETITION

Optimizing Client Segmentation and Credit Risk Strategies

Team: Group 3



Koneru Harshitha
MSBA, Fall 24 Cohort



Belle He
MSBA, Fall 24 Cohort



Sri Teja Adimulam
MSBA, Fall 23 Cohort



Yizhi Chen
MSBA, Fall 24 Cohort

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Key Takeaways

DATASET OVERVIEW

36,000+

Records

0-60

**Months Payment
History**

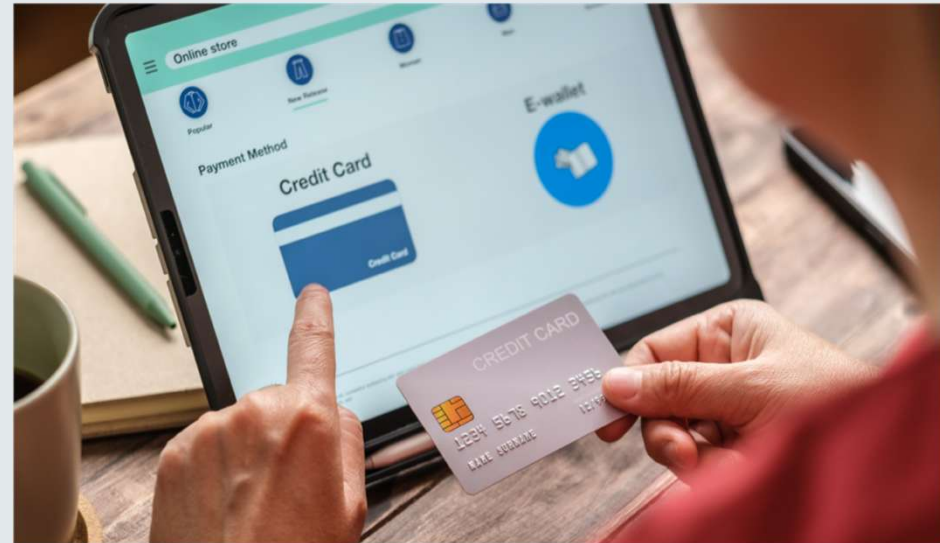
28+

Variables

DATASET STRUCTURE

KEY VARIABLES CLASSIFIED

- **Demographics:** Age, Gender, Family Size, Marital Status
- **Employment:** Education Type, Income Type, Annual Income, Occupation Type
- **Housing:** Property Ownership, Housing Type, Car Ownership
- **Payment History:** Repayment Patterns, Overdue Status, Client Classifications (Good/Bad)



DATA CLEANING & PREPROCESSING



Variables EDUCATION_TYPE, MARITAL_STATUS, and HOUSING_TYPE were excluded due to low sampling adequacy based on the Chi-Square Test

Variable Exclusion

Encoding Categorical Variables

NAME_INCOME_TYPE was converted into dummy variables representing each income category.



All columns were converted to numeric format for algorithm compatibility.

Standardization & Numeric Conversion

Correlation Analysis

A correlation matrix was created to detect multicollinearity among numeric variables.



STATISTICAL ANALYSIS

Naïve Bayes

Confusion Matrix of Naive Bayes			
	0	1	
0	1367	11	
1	7673	63	

Predictive Accuracy = 15.69%



Logistic Regression Model

Confusion Matrix of glm model			
	0	1	
0	895	483	
1	4797	2939	

Predictive Accuracy = 42.07%



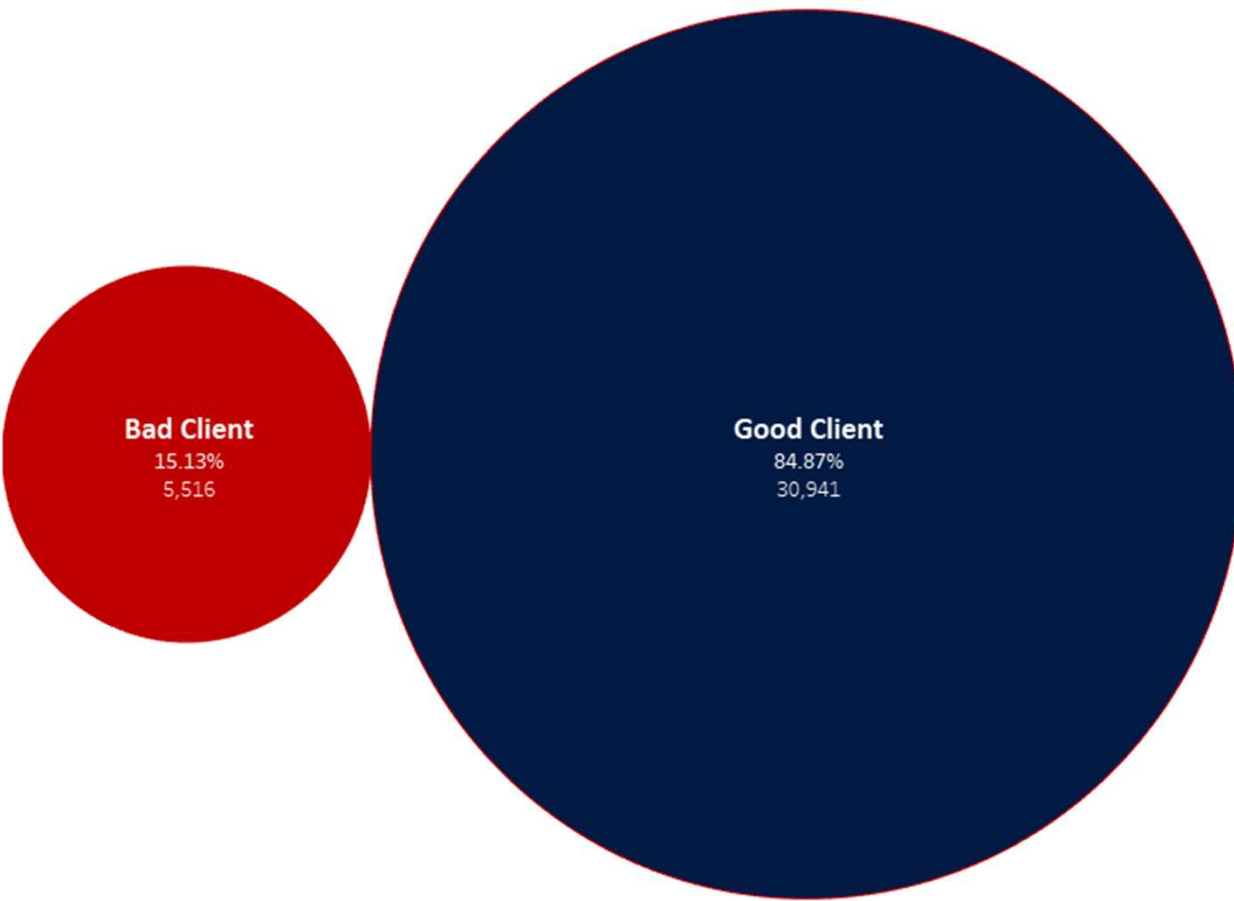
Decision Tree Model

Confusion Matrix of Decision Tree			
	0	1	
0	19	1359	
1	14	7722	

Predictive Accuracy = 84.94%



TABLEAU EXPLORATORY ANALYSIS



Addressing the Obvious!

This distribution is the same across all variables such as

- Demographics
- Employment
- Housing, etc

Income Distribution of Clients

- Average Annual Income of Clients stands at 185,504.
- Major Clients fall within lower income ranges
- Few outliers significantly exceed the average
- A small proportion of clients exceed 1M, showing high-income outliers.

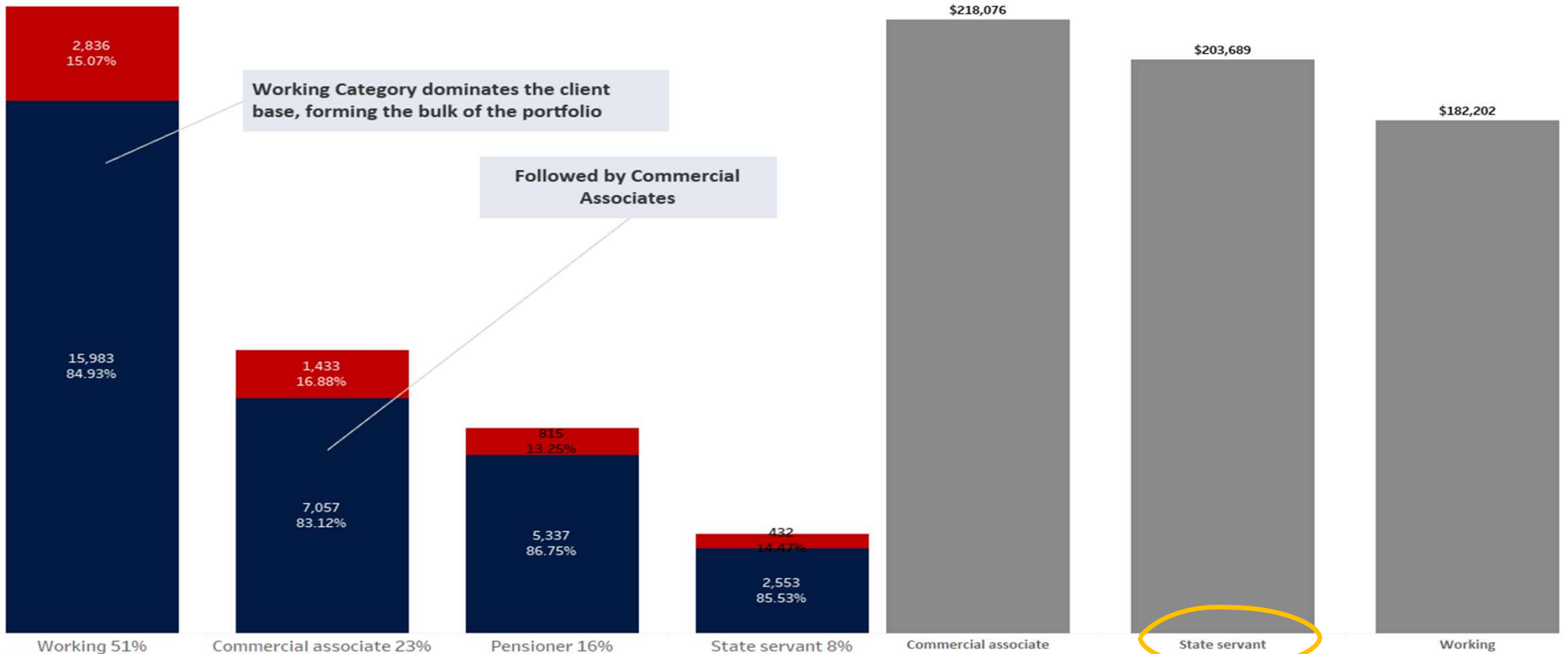
Upper Whisker:	378,000
Upper Hinge:	225,000
Median:	157,500
Lower Hinge:	121,500
Lower Whisker:	27,000



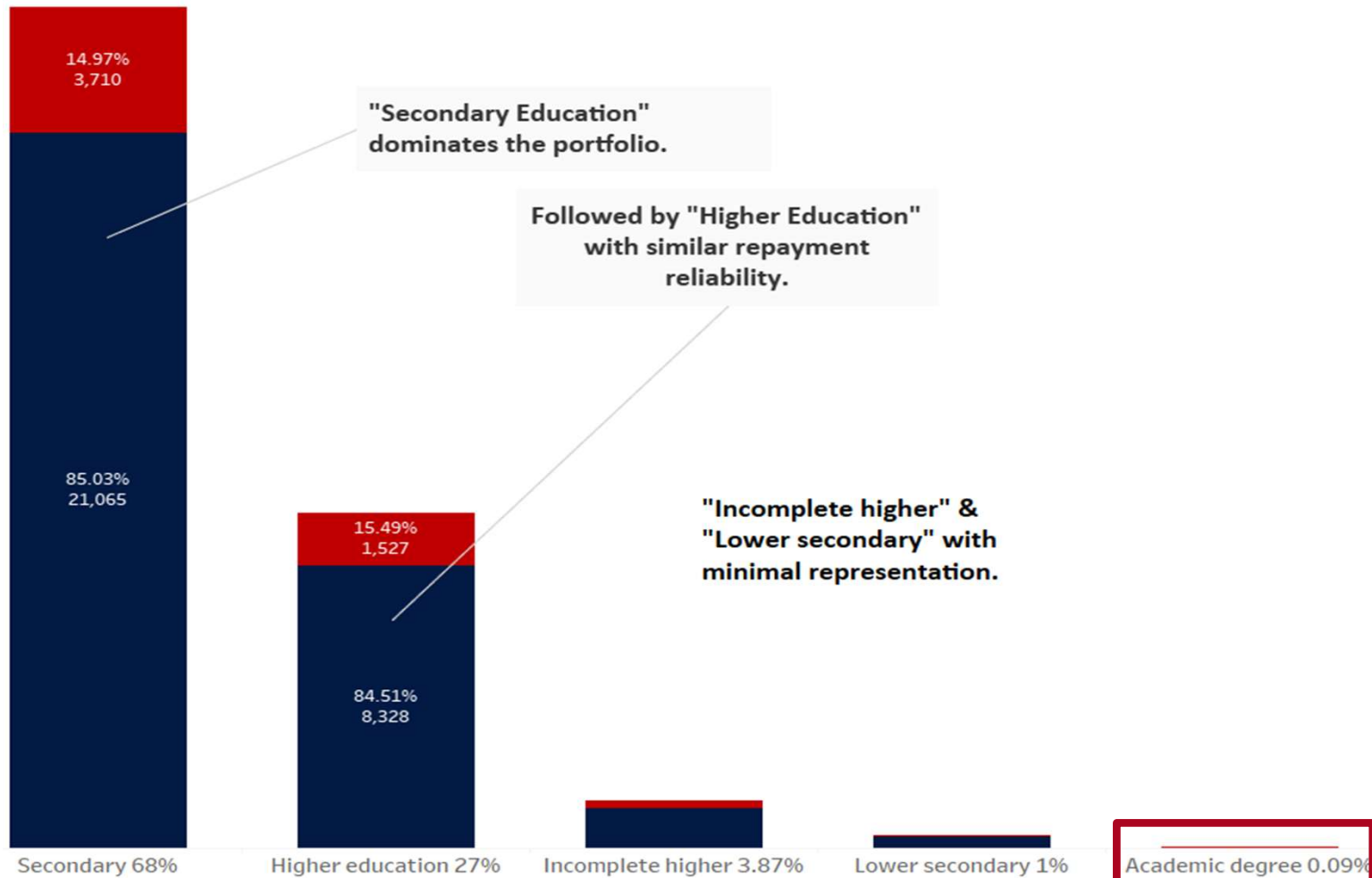
Insight 1: Income Type Spread

Client Distribution by Income Type

Average Income by Income Type



Insight 2: Education Type Spread



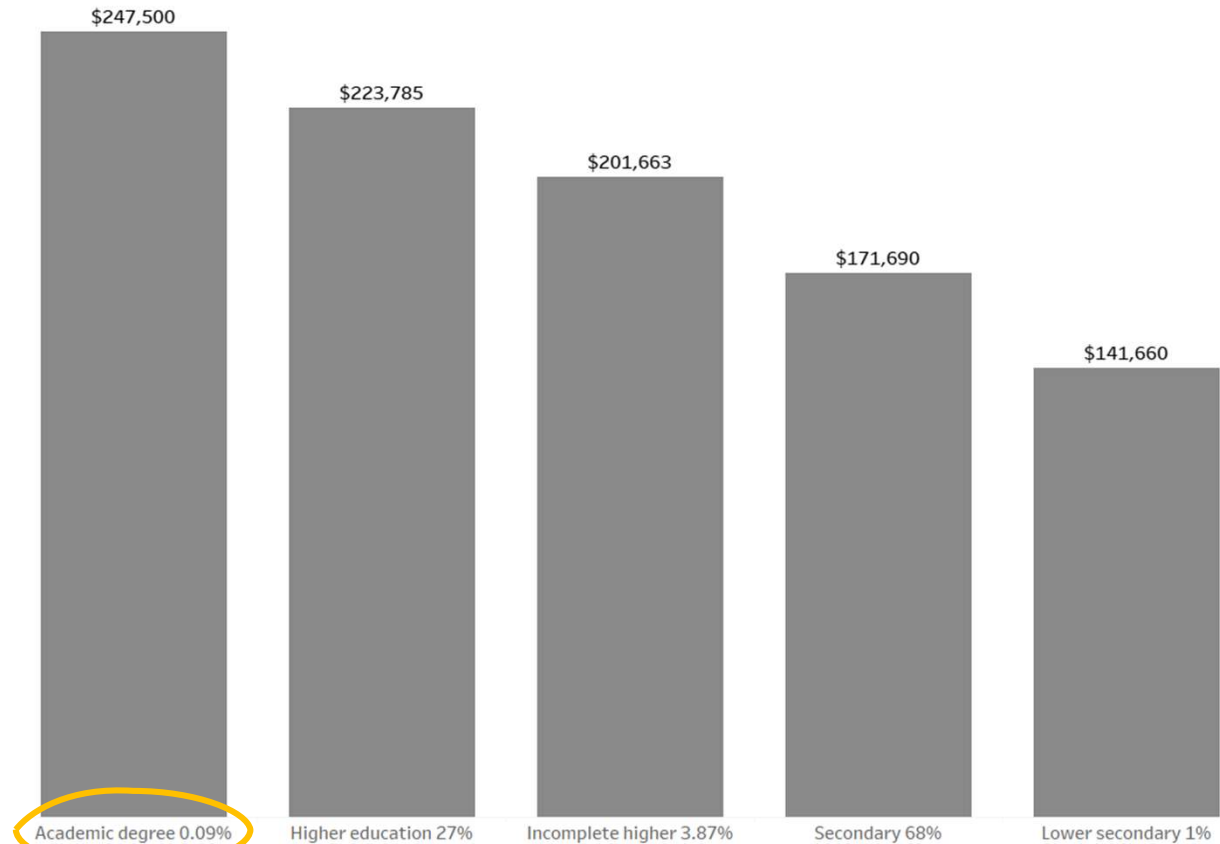
Academic Degree Client Segment

Academic Degree Distribution



Tiny group but exceptional repayment reliability.

Average Income by Education Type

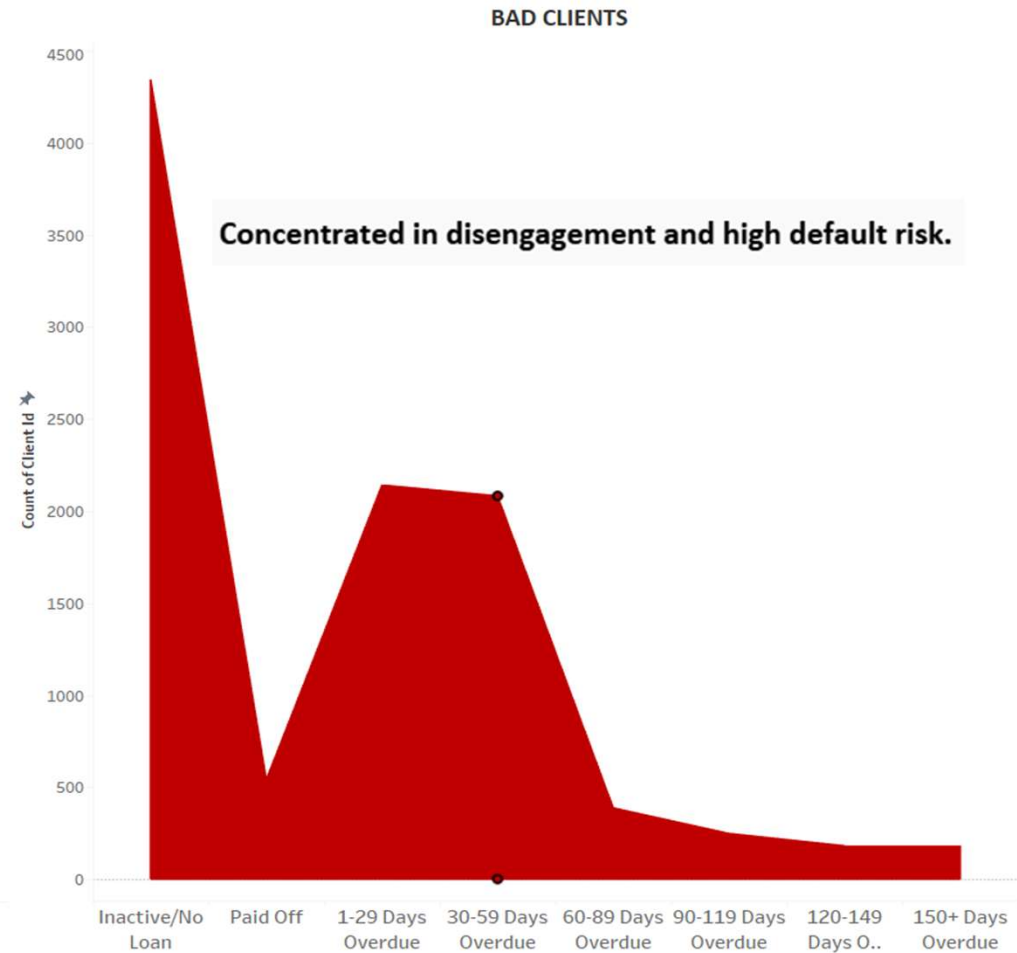
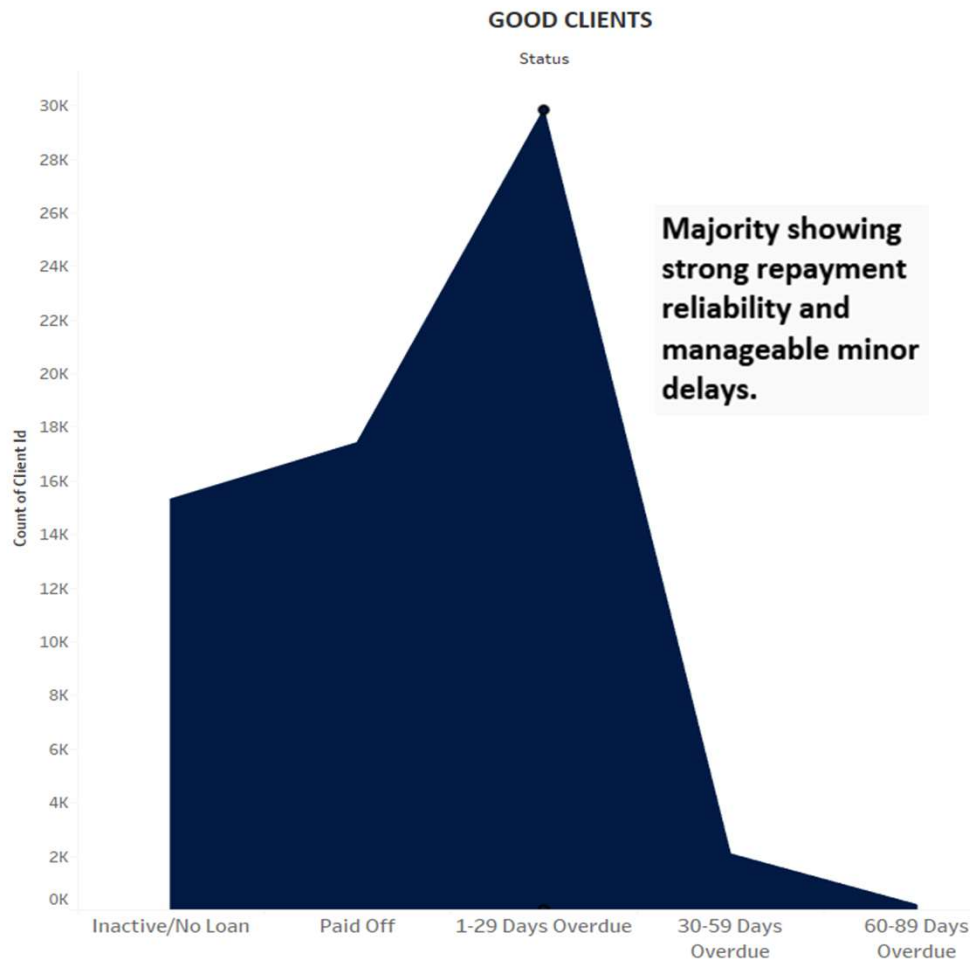


Insight 3: Credit Record

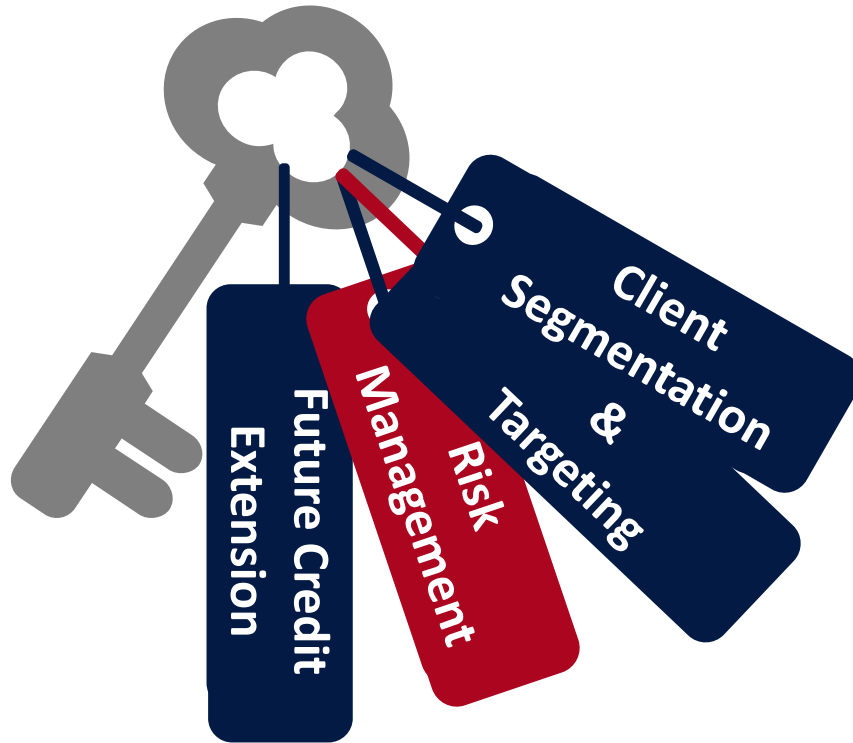
Credit Repayment Behavior based Client Classification

	Inactive/No Loan	Paid Off	1-29 Days Overdue	30-59 Days Overdue	60-89 Days Overdue	90-119 Days Overdue	120-149 Days Overdue	150+ Days Overdue
Good Client	15,302	17,409	29,822	2,067	168			
Bad Client	4,346	541	2,141	2,082	387	248	181	180

Insight 3: Credit Behavior



RECOMMENDATIONS



Develop tailored strategies to identify and engage specific client groups, optimizing portfolio composition and profitability.



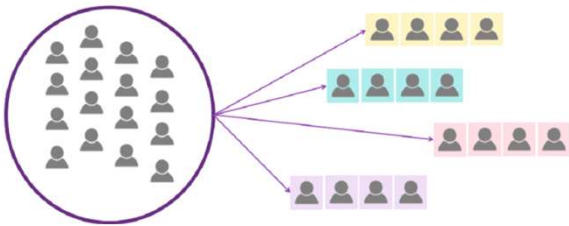
Implement proactive measures to mitigate credit risk and address high-risk client behaviors to enhance portfolio stability.



Leverage insights to scale credit operations efficiently, focusing on reliable and high-potential client segments for sustained growth.

Client Segmentation & Targeting

Optimizing Client Groups for Portfolio Growth

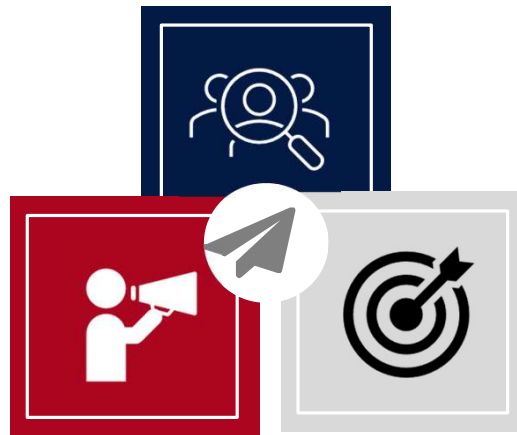


Focus on Low-Risk Groups

Develop targeted financial products for Pensioners and State Servants, leveraging their reliability and low-risk profiles.

Optimize Marketing Efforts

Prioritize Working Clients for scaling efforts while nurturing high-potential segments with specific campaigns.



Target Underrepresented Segments

Introduce tailored credit plans for Students and Academic Degree Holders to grow their representation.

Risk Management

Mitigating Risk Through Proactive Strategies

01

Early Warning Systems



Proactively monitor clients in 1–29 Days Overdue and 30–59 Days Overdue categories to prevent escalation

02

Focus on High-Risk Groups



Address 90+ Days Overdue clients with debt restructuring or settlement programs

03

Credit Scoring Adjustments



Penalize repeated Inactive/No Loan behavior in scoring models to ensure responsible credit extension

Future Credit Extension



Scaling Operations for Growth

→ Scale Operations for Reliable Segments



Expand credit for Working Clients and Secondary Education Clients, who dominate the portfolio

→ Target Premium Products



Focus on Academic Degree Holders and high-income clients with tailored premium offerings

→ Flexible Repayment Plans



Design custom repayment options for Pensioners and State Servants to encourage portfolio growth

Questions!



Thank You

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Recommendations

Client Segmentation & Targeting

Risk Management

Future Credit Extension

Questions!

